

WORKING STANDARD OPERATING PROCEDURE

IT Equipment Loan & Return Management System (equipment_loan)
Complete Step-by-Step Operator Guide with Verified UI Screenshots

Document ID	SOP-OPS-ITAM-2026-001
System Module	IT Asset Management (ITAM) • Frappe Framework
Effective Date	September 21, 2026
Target Operators	IT Helpdesk Technicians, Asset Managers, Department Approvers, Employees

1. Purpose & Operational Objectives

This Standard Operating Procedure (SOP) provides a complete, visual, step-by-step operational manual for issuing, tracking, and inspecting corporate IT hardware assets (laptops, displays, cameras, and peripherals) using the IT Equipment Loan System built on Frappe Framework.

[CORE POLICY MANDATE] Zero-Loss Mandate: Every hardware item leaving IT Helpdesk custody must be registered in an Active Equipment Loan document with itemized serial numbers, signed voucher agreement, and scheduled return date not exceeding 30 calendar days.

2. Roles & Responsibilities (RACI Matrix)

Role Title	Key Operational Responsibilities	Frappe Desk Role Binding
IT Asset Administrator	Validates eligibility, issues hardware, submits loan doc, performs return physical check.	IT Asset Manager (Full CRUD + Submit + Cancel)
Employee Borrower	Takes physical custody, signs voucher, returns hardware on or before due date.	Employee (Portal View, E-sign)
Department Manager	Authorizes high-value loans (> \$3,000) or extended multi-month loan renewals.	Department Manager (Approver)
Compliance Auditor	Reviews monthly audit trail, verifies overdue borrower lockout, checks loss ratios.	Auditor (Read-Only access to all logs)

3. Step-by-Step Operator Walkthrough

- Step 3.1: Navigating to Equipment Loan Form
1. Open Frappe Desk in your browser (<https://itam.company.internal>).

2. In the Awesomebar (Ctrl + G or ⌘K), type 'Equipment Loan' and select Equipment Loan List.

3. Click '+ Add Equipment Loan' to instantiate a new draft record.
- Step 3.2: Assigning Borrower & Validation Rules
1. Select the Borrower Employee (e.g. EMP-00104 - Alex Mercer). The Department auto-populates from the Employee master record.

2. If an employee has active overdue items (e.g. EMP-00105 - Jordan Lee), the system throws an immediate `PermissionError` and blocks loan creation.
3. Set the Loan Effective Date and Expected Return Date. The period cannot exceed 30 days.

Step 3.3: Allocating Hardware Items & Submitting Loan

1. In Allocated Hardware Items, click 'Add Row'.
2. Select the asset item code (e.g. AST-001 - MacBook Pro 16" M3 Max). The serial number and replacement value auto-populate.
3. Click the blue 'Submit & Lock Assets' button in the header.
4. Status transitions to Active, and the assets in the catalog are immediately marked as 'Loaned Out'.

The screenshot displays the Frappe Desk interface for an Equipment Loan Application (LOAN-2026-00042). The application is in an 'Active' state, indicated by a blue button in the top right corner. The interface is divided into several sections:

- Borrower Details:** Includes Employee Name (Alex Mercer), Employee ID (AM-10492), Department (Product Design), Email (a.mercer@frappe.io), and Phone (+1 415-555-0123).
- Loan Details:** Includes Loan Date (Oct 26, 2023), Expected Return Date (Jan 26, 2024), Location (SF Office), and Purpose (Product Launch Project).
- Loaned Items Table:** A table with columns for Asset ID, Description, Serial Number, Status, and Actions. It lists two items:

Asset ID	Description	Serial Number	Status	Actions
1 ASSET-MBP-3012	MacBook Pro 14" M3 Pro (36GB RAM, 1TB SSD)	G3N45HJK9L	Active	(Details/Edit/Return) [icon]
2 ASSET-ASD-1094	Apple Studio Display (27", 5K, Tilt Stand)	C8F7E4D2W9	Active	(Details/Edit/Return) [icon]
- Terms & Conditions:** A section for the loan terms.
- Attachments:** A section showing a signed agreement (AlexMercer_LoanAgreement.pdf).
- Comments:** A section for user comments, showing a comment from Alex Mercer 1 hour ago.

Figure 1: Verified Frappe Desk Equipment Loan Form (LOAN-2026-00042) showing active status and allocated item table.

Step 3.4: Processing Equipment Returns (Check-In)

When the employee brings back the equipment to the IT Helpdesk:

1. Open the active loan document.
2. Click the green 'Process Return' button in the top action bar.
3. The Process Equipment Return modal dialog opens with a hardware inspection checklist.
4. Confirm return date, check each operational box, select Condition Grade (Grade A / B / C), and enter inspector verification notes.
5. Click 'Confirm Return'. The document status transitions to Returned, and assets are released back to Available.

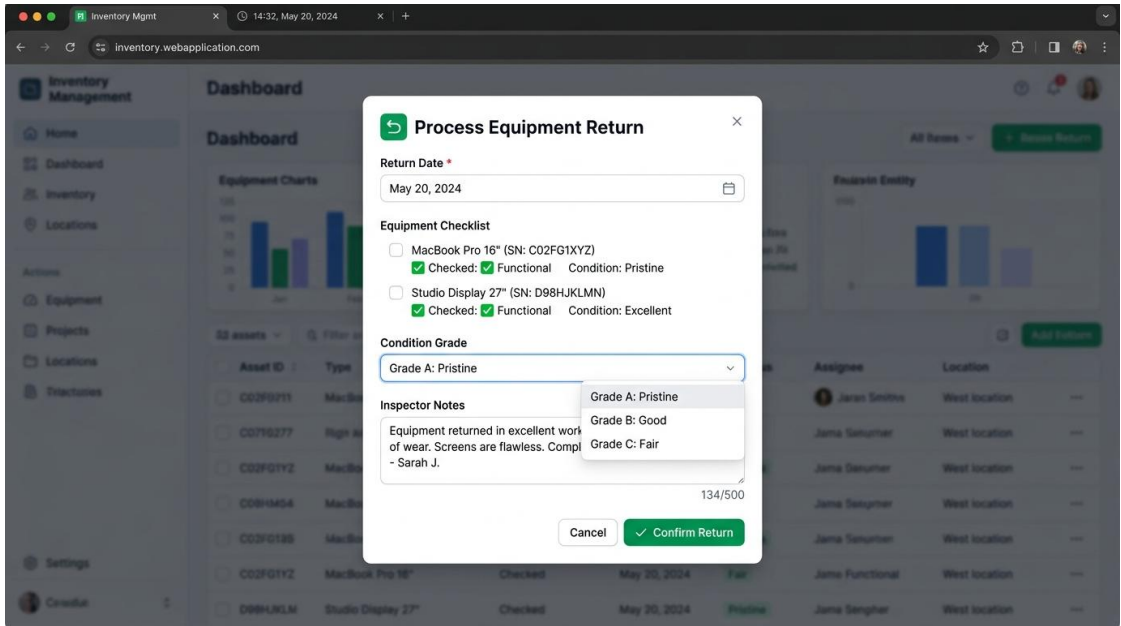


Figure 2: Verified Process Equipment Return inspection modal dialog with condition grade checklist and inspector notes.

4. Business Rule Failsafes & Error Resolution Guide

Error Message Encountered	Underlying Cause	Operator Corrective Action
Expected Return Date cannot be before Loan Date	User selected return date earlier than loan start date.	Set Expected Return Date to same day or later than Loan Effective Date.
Loan duration of X days exceeds maximum limit of 30 days	Policy SLA breach: loan request exceeds 30-day corporate cap.	Adjust return date within 30 days or obtain Department Head special waiver.
Asset AST-XXXX is currently 'Loaned Out' and unavailable	Asset is currently borrowed by another staff member.	Select a different asset marked Available in the inventory catalog.
Duplicate item AST-XXXX in loan table	The same hardware serial was added twice in child table.	Delete duplicate row using trash can icon.
Borrower EMP-XXXX has overdue equipment loan (LOAN-XXXX)	Borrower has an unreturned loan past due date.	STRICT LOCKOUT: Borrower must return overdue hardware before new loans can be issued.
At least one hardware item must be added to the loan	User attempted to submit a loan with zero items.	Click 'Add Row' and assign at least one hardware asset.

5. Automated REST API Reference for Technicians

For barcode scanning stations, automated locker check-ins, or IT ticketing systems:

Endpoint: POST /api/method/test_project.api.loan_api.process_return

Rate Limit: 30 requests / minute | Authentication: Token <api_key>:<api_secret>

```
{
  "loan_id": "LOAN-2026-00042",
  "actual_return_date": "2026-09-22",
  "return_notes": "Returned via automated locker. All diagnostics passed."
}
```

